Note on MLS eqn. (2.37) p. 42

MLS eqn. (2.37) should not be interpreted as a composition of two rigid displacements, as in eq. (2.24) p. 37, even though it has a similar form, since $e^{\hat{\xi}\theta} \in SE(3)$ (proposition 2.8, p. 41). As stated, $g_{ab}(0)$ is the initial configuration,

$$g_{ab}(0) = \begin{bmatrix} R_{ab}(0) & p(0) \\ 0 & 1 \end{bmatrix} ,$$

not really to be thought of as an initial displacement. So, as the rigid body associated with frame B displaces,

$$\begin{aligned} g_{ab}(\theta) &= e^{\hat{\xi}\theta} g_{ab}(0) = \begin{bmatrix} e^{\hat{\omega}\theta} & M(\theta, v) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{ab}(0) & p(0) \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} e^{\hat{\omega}\theta} R_{ab}(0) & e^{\hat{\omega}\theta} p(0) + M(\theta, v) \\ 0 & 1 \end{bmatrix} , \end{aligned}$$

where

$$M(\theta, v) = \left(I - e^{\hat{\omega}\theta}\right) \left(\hat{\omega} + \omega\omega^T \theta\right) v \, ,$$

a submatrix of the matrix in eqn. (2.36). After the concept of a screw is defined, and interpretted geometrically as a twist (proposition 2.9 p. 42), we see that this $M = p$ (eqn. (2.38) p. 43), the displacement associated with the screw or the exponential mapping of the twist ξ, $e^{\hat{\xi}\theta}$.

So, as the rigid body displaces, its rotation matrix changes to be $e^{\hat{\omega}\theta} R_{ab}(0)$ and the coordinates of the body fixed frame changes to be $M(\theta, v)$ plus the rotated initial displacement vector, $e^{\hat{\omega}\theta} p(0)$.